

RapidBoxForest: Link-Envelope-Guided Box Forests for Fast Multi-Query Manipulation

TIAN, Yu and Ren, Hongliang

Abstract—Fast repeated-query manipulation benefits from reusable volumetric planning structure, but high-quality IRIS-style convex regions are expensive to construct and rebuild. This paper presents RapidBoxForest (RBF), a link-envelope-guided box-forest planner for fast multi-query manipulation. The central geometric primitive maps endpoint interval envelopes to link interval envelopes via capsule Minkowski decomposition and convex-hull generator bounding applied to *C-space box validation and multi-query reuse*: endpoint interval envelopes bound rounded-link endpoint motion over a joint box, and a radius-lifted link envelope provides an inexpensive filter for box validation and collision rejection. A rapidly-exploring random tree (RRT)-inspired grower builds a scene-specific box forest into a queryable corridor graph, and a Lifelong Envelope Cache Tree (LECT) selectively caches kinematics-keyed envelope records when replay is cheaper than recomputation. On the canonical Shelf+IIWA workload and on balanced IIWA, UR5, and Panda easy/medium/hard random scenes, RBF-SH achieves reusable build times of 0.15–1.1 s and online query times below 0.11 s, with final-audited path lengths competitive with sampling-based baselines at a fraction of the reusable build cost. Secondary mechanism studies quantify endpoint-source and link-envelope representation trade-offs and show that LECT prewarm provides measurable cross-scene gains. The results position link-envelope-guided box forests as a practical multi-query planning system with an optional conservative corridor validation mode. The accompanying artifact provides the full implementation and reproduction instructions.

Index Terms—Motion planning, multi-query planning, link interval envelopes, box-region planning, final audit.

I. Introduction

Motion planning for high-degree-of-freedom (DOF) manipulators becomes especially demanding when a system must answer many related queries rather than one isolated instance. In warehouses, packing cells, and shared workcells, robot kinematics stay fixed while obstacles, task endpoints, or local scene edits change. The central question is how to construct reusable *C-space* planning structure quickly enough that repeated queries can benefit from region-level search without paying a large scene-specific preprocessing cost.

Iterative Regional Inflation by Semidefinite Programming (IRIS) and Graph-of-Convex-Sets (GCS) methods show why free-region construction is valuable: once good convex regions are available, graph search or convex optimization can produce high-quality motions over those regions [1], [2], [3]. The limitation is construction cost. In high-dimensional manipulator scenes, inflating and validating high-quality convex regions can dominate the planning budget and must often be repeated when the scene changes. Sampling-based planners such as probabilistic roadmap (PRM) and rapidly-exploring random tree (RRT) methods avoid this region-construction cost [4], [5], [6], [7], but their reusable structures are usually zero-volume graphs, so repeated queries still pay for local collision checking and do not obtain a reusable free-region cover.

RBF targets a different design point by constructing simpler box regions quickly. Instead of inflating general polytopes, it grows axis-aligned boxes in configuration space, connects accepted boxes into a graph, and answers later queries by searching the resulting box corridor. This trades per-region expressivity and sometimes path length for cheap construction, simple adjacency, and direct reuse across many queries. The intended claim is a system-level one: a fast box-region planning pipeline can be a useful final-audited alternative when multi-query latency matters more than obtaining the shortest path from a more expensive region decomposition.

The key technical ingredient is a link interval envelope computed from endpoint interval envelopes. For rounded link models, RBF reduces link motion over a joint box to the motion of a zero-radius centerline plus a final radius lift. Endpoint interval envelopes bound the endpoint trajectories over the joint box; link interval envelopes then enclose the swept rounded link and can be stored as axis-aligned bounding boxes (AABBs), *k*-DOP slabs, or SupportHull records. Strict interval sources provide an optional conservative validation path, while tighter practical sources are used as fast filters and discharged by the same final scene audit used for all reported planners.

On top of this geometric primitive, the grower borrows the rapid frontier-expansion idea of RRT: samples choose useful frontiers, a local region oracle materializes a free box around the projected seed, and accepted boxes extend a connected box forest. The result is also an adaptive cell-decomposition view of manipulator *C-space*,

but one biased toward the regions needed by the current repeated-query workload. A Lifelong Envelope Cache Tree (LECT) then memoizes scene-independent endpoint and link-envelope evidence keyed by robot kinematics and joint intervals; it is an acceleration layer for repeated materialization, not a separate planning claim.

The evaluation therefore asks where this envelope-based box forest lies on the final-audited time/path-quality frontier of the implemented pipelines. The main axes are reusable build time, online query time, fixed-resolution audited success, path length, amortization, and scene/robot variation. The conservative box-corridor statement is kept separate from the main comparisons because it requires both strict validation and path containment in the validated box union, whereas the reported paths may be changed by the common post-processing step. Endpoint/link envelope studies and LECT replay are reported to explain construction cost, with LECT treated as an optimization layer. IRIS-NP+GCS is used as an important pipeline baseline under the same workload-level seeds used by the other planners: manually selected reference anchors in the Shelf+IIWA scene and fixed random seeds in the random-scene study.

We therefore position the paper as a planning-systems contribution enabled by a geometry primitive. The endpoint-to-link envelope is the main geometric idea; the planner and artifact show how that primitive supports a reproducible low-build-cost multi-query design point under final audit.

The paper’s main contribution is a reproducible final-audited multi-query planning system and its measured design point. It is supported by three concrete elements:

- 1) *An endpoint-to-link envelope primitive for C-space box validation.* Capsule Minkowski decomposition and convex-hull generator bounding are individually classical facts already combined in articulated-body CCD work [8], [9], [10]. RBF applies this two-step pipeline to *C-space box growth and multi-query reuse*: a switchable endpoint source family—AA-backed interval forward kinematics (IFK), hierarchical IFK (HIFK), CritSample, analytical extrema—compact per-representation link records (AABB, KDOP26, SupportHull) for staged filtering, and a kinematic-keyed LECT cache for cross-scene envelope reuse. No prior work has been found applying this pipeline to *C-space box validation or multi-query planning reuse* (section III and tables III and IV).
- 2) *A self-contained link-envelope-guided box-forest planner.* Scene-specific *C-space box forests* are grown through seeded frontier expansion, consolidated into a connected corridor graph, and queried by graph search and local refinement. LECT provides an optional kinematic-keyed cache for repeated envelope materialization across builds (sections IV and V and table V).
- 3) *A scoped final-audited design-point evaluation.* Shelf+IIWA and balanced IIWA/UR5/Panda scenes

are evaluated under common post-processing and fixed-resolution final audit. RBF achieves low reusable build/query cost with near-baseline audited path lengths; path quality improvements over the strongest baselines and grower-strategy ablations remain future work (figs. 3 and 4 and tables I and II).

The paper’s evidence chain is: Section III develops the endpoint and link interval-envelope primitive for fast box validation; section IV describes the cache optimization that reuses expensive envelope records; section V turns link-envelope-filtered boxes into a rapidly expanding box-forest planner; and section VI evaluates the resulting final-audited build/query and path-quality trade-offs as implemented pipeline design points with IRIS-NP+GCS, PRM, RRTConnect, and BIT* as contextual baselines.

II. Related Work

A. Convex Region Planning and Graphs of Convex Sets

Convex region planners approach motion planning by constructing a certified cover of free space and then solving graph search or convex optimization on top of that cover. IRIS introduced the now-standard idea of computing large collision-free convex regions by alternating separating hyperplanes and region inflation [1]. Later work extended the same line into configuration space, improved region quality, or improved graph compactness through visibility-graph and clique-cover constructions [11], [12], [13]. Marcucci *et al.* made the Graph-of-Convex-Sets formulation particularly influential by showing that motion planning can be cast as optimization over a graph of certified convex regions [2], [3].

RBF is aligned with this family in objective but not in geometric primitive: it uses axis-aligned boxes in *C-space* rather than general polytopes, trading per-region expressivity for lower build overhead and simpler adjacency bookkeeping. The resulting cover is coarser than a high-quality convex decomposition, but it avoids repeated hyperplane-based region inflation and is easier to rebuild under repeated-query workloads. The comparison in this paper therefore concerns fast reusable free-region construction versus tighter, more scene-coupled regional convexification, rather than box and polytope representations in isolation.

B. Swept Volume Approximation and Capsule Geometry

Computing the swept volume of a moving rigid body has been studied extensively in computational geometry and computer-aided manufacturing (CAM) communities [14], [15]. For a convex body $P = \text{conv}(v_1, \dots, v_m)$, bounding each generator trajectory by a conservative outer set $\hat{\Gamma}_i(I)$ yields

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right),$$

a bound that is implicit in oriented bounding box (OBB) tree construction for swept mesh primitives [16].

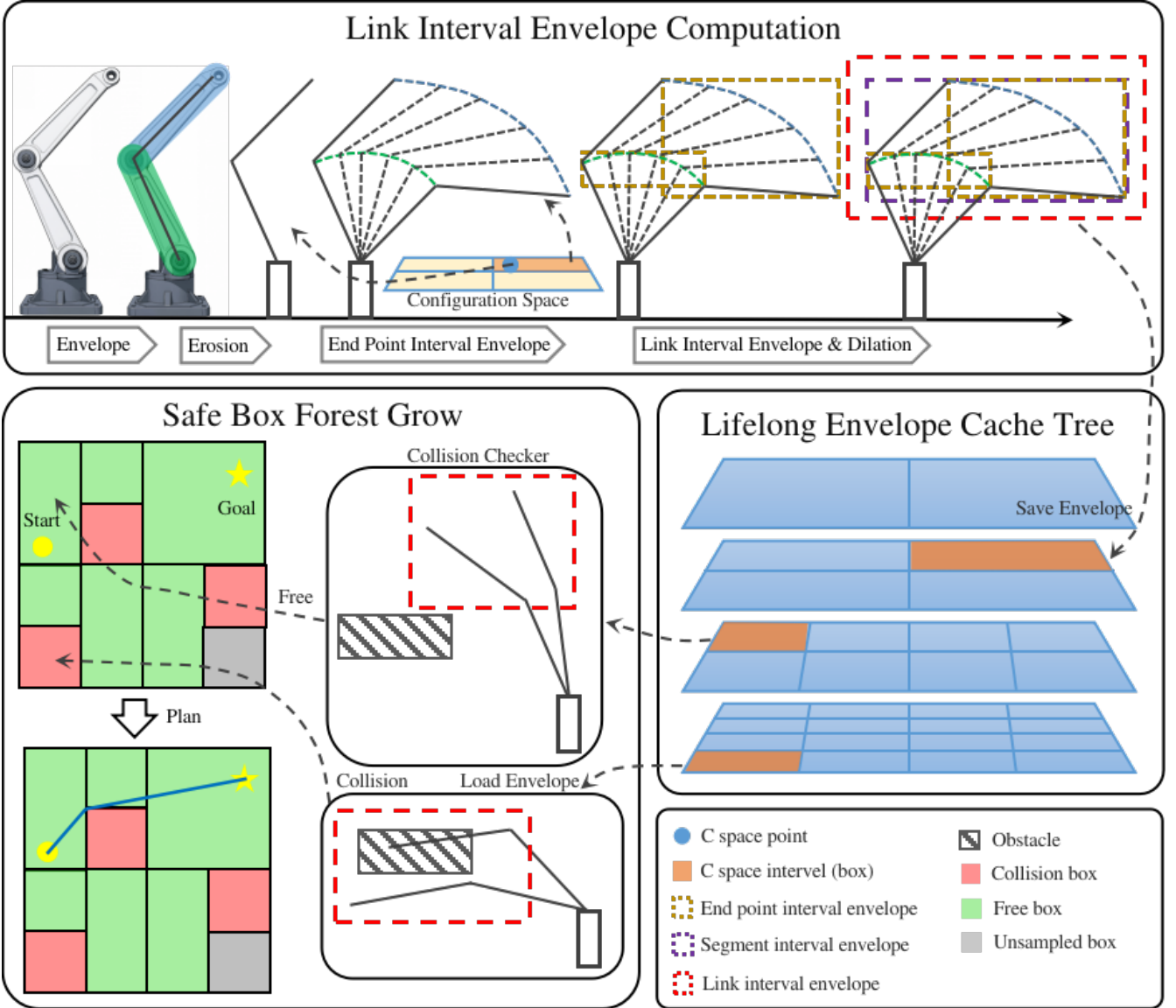


Fig. 1. RBF pipeline for link-envelope-guided multi-query planning. Top: endpoint interval envelopes induce link interval envelopes over a C -space box. Middle: LECT optionally reuses kinematics-keyed endpoint and link-envelope records across repeated builds. Bottom: scene filtering and an RRT-inspired grower expand a connected box-region forest for lightweight corridor queries and final audit.

For capsule links this bound reduces to bounding only the two segment endpoints. The Proximity Query Package (PQP) exploits exactly this Minkowski decomposition ($C = S \oplus B_2(r)$, sweep of C reduces to sweep of S plus radius lift) [17]; GJK and its robust implementation by van den Bergen apply the same identity for distance queries [18], [19]; the Flexible Collision Library (FCL) generalises the swept-sphere interface to a full collision library [20].

C. Interval Forward Kinematics and the CCD Endpoint-Bounding Pipeline

Interval arithmetic [21], [22] provides inclusion-monotone enclosures of real-valued functions over parameter intervals. Snyder applied interval arithmetic to parametric geometry to compute conservative bounding boxes for curves and surfaces without mesh sampling [23].

Merlet extended DH-chain interval propagation to certified workspace analysis for parallel and serial manipulators [24], [25].

These ideas meet directly in the continuous collision detection (CCD) literature for articulated bodies. Redon, Kim, Lin, and Manocha [8] model each robot limb as a line-swept sphere (LSS, i.e. capsule) and propagate interval Denavit-Hartenberg (DH) transforms along the kinematic chain to obtain *interval vectors bounding the two LSS endpoint positions* over the motion time step; the swept capsule is then bounded by padding the endpoint bounding box with the link radius. This is explicitly the two-step pipeline

$$\begin{aligned} & \text{interval FK endpoint bounding} \longrightarrow \\ & \text{bbox}([\mathbf{p}_i] \cup [\mathbf{p}_{i+1}]) \oplus B_2(r) = \text{link swept-capsule bound} \end{aligned}$$

that is the geometric core of RBF. The journal extension by Redon, Lin, Manocha, and Kim [9] applies the same

framework to general robot arms and kinematic trees; that work has accumulated over 240 citations and is a standard CCD reference. Zhang *et al.* [10] replace interval arithmetic with Taylor models in the same pipeline to reduce wrapping errors, analogous to how RBF’s CritSample and analytical sources tighten the IFK bound. Schwarzer *et al.* [26] address a multi-query motivation similar to RBF’s—reusing collision evidence across planning calls—but at the path-segment level rather than as volumetric C -space records. Corrales *et al.* [27] apply sphere-swept line (SSL) bounding volumes dynamically for human–robot safety monitoring using the same structural observation that bounding a capsule’s motion reduces to bounding its two endpoints.

D. Novelty Boundary: Prior CCD vs. RBF

The two-step endpoint-interval-to-swept-capsule pipeline is therefore unambiguously present in the CCD literature for time-step motion. The differences that motivate the RBF contribution claim are: (a) the parameter domain is a C -space *joint-box interval* rather than a trajectory time step, so the goal is to classify whether *any* configuration in the box is collision-free, not whether a single given path collides; (b) the endpoint source is a switchable interface—IFK (conservative), CritSample (practical), and analytical extrema (tightest)—rather than a fixed algorithm, allowing the conservative vs. performance trade-off to be controlled without changing the link-representation or planner layers; (c) compact per-representation link records (AABB, KDOP26, SupportHull) are organised for staged filtering during box growth, a knob absent from CCD libraries; and (d) the endpoint and link envelope records are organised in LECT, a persistent *kinematic-keyed* cache that reuses evidence across multi-query builds in related scenes, a reuse level not addressed by CCD libraries or path-segment caches. No prior work has been found that applies the two-step endpoint bounding pipeline to C -space box growth or to multi-query volumetric planning reuse. RBF is complementary to the CCD stack: link envelopes reduce repeated geometric work on candidate boxes, while discrete scene checks remain the final authority for the reported paths.

E. Multi-Query Reuse and Adaptive Configuration-Space Decomposition

Multi-query planners typically amortize effort by reusing a graph structure such as a roadmap. PRM is the classical example: build a connectivity graph once and reuse it for many start-goal pairs [6], [28]. LazyPRM, lazy edge checking, and certificate-based planners postpone or cache edge validation so that online search does not repeatedly evaluate the same discrete local connections [29], [30]. Experience-based and trajectory-library systems reuse prior solutions or local motion snippets, often as initialization or bias for a new query

rather than as a conservative decomposition of a region of configuration space [31].

RBF operates at a different reuse level. It grows and reuses volumetric boxes rather than only paths, binary collision outcomes, or zero-volume edges. PRM-like methods reuse connectivity and must revalidate affected edges after scene changes. Convex-region planners reuse higher-quality regions, but those regions are usually coupled to the obstacle layout in which they were constructed.

The method also inherits the central idea of adaptive cell decomposition and subdivision planning: represent free space by cells, split cells where the current evidence is insufficient, and search the adjacency graph induced by the accepted cells [32], [33]. The difference is the operating regime. Classical cell-decomposition examples are often low-dimensional workspace or explicit C -space constructions, whereas RBF applies the same adaptive-cell idea to high-dimensional manipulator C -space through a link-envelope oracle. Scene-specific obstacle filtering is layered on top of reusable kinematic envelope evidence, with LECT available only as a cache for expensive records. The trade-off is explicit: RBF accepts axis-aligned boxes and over-approximation in exchange for simple validation, adjacency maintenance, and repeated-query throughput.

F. Sampling-Based Planning and Trajectory Optimization

Sampling-based planners remain standard baselines when the goal is to solve a single query quickly. RRT-Connect is particularly effective in manipulator planning because bidirectional growth often succeeds with very little global structure [34]. Asymptotically optimal and informed variants such as RRT*, Batch Informed Trees (BIT*), and Fast Marching Tree (FMT*) improve path quality and search efficiency when more time is available [7], [35], [36], [37]. The Open Motion Planning Library (OMPL) has made these methods easy to deploy and benchmark in a common software stack [38], [39]. However, standard RRT variants produce single-use zero-volume evidence and typically restart from scratch on the next query.

By contrast, RBF uses rapidly exploring, adaptive expansion to sample and materialize C -space boxes from the envelope oracle, producing a volumetric corridor structure that can be queried repeatedly within the same environment. Trajectory optimizers address a different stage of the pipeline. CHOMP, STOMP, TrajOpt, and Gaussian-process motion planners refine an initial path by local descent or sequential convex optimization [40], [41], [42], [43]. RBF uses these methods as complementary points of comparison: its aim is to provide a reusable volumetric backbone on which graph search and local post-processing can operate.

III. Link Interval Envelopes

This section develops the main geometric primitive used by RBF: a map from a manipulator joint box to compact link interval envelopes. The goal is to make

box validation cheap enough for repeated-query planning. A strict interval source gives a conservative validation option, while tighter sampled or support-based variants are practical filters whose reported paths are discharged by the common final audit. Thus the strongest free-region statement in this section is conditional on the conservative source and on paths remaining inside the validated box union; the main planning experiments report final-audited performance separately. The construction first bounds endpoint trajectories and then applies a constant-radius lift for rounded link geometry.

Here $\mathcal{Q}$ denotes the joint-limit box explored by RBF, $I \subseteq \mathcal{Q}$ one joint interval, and $\mathcal{O} \subset \mathbb{R}^3$ the obstacle set. Endpoint envelope $E_k(I)$ outer-bounds endpoint k , and link envelope $B_\ell(I)$ outer-bounds the swept volume of link ℓ over the same interval. Together they define the reusable planning primitive $I \mapsto \{E_k(I)\} \mapsto B_\ell(I)$. When the endpoint source is itself conservative, the resulting link envelope is conservative as well; when a tighter empirical source is used, the same representation serves as a fast broad- or mid-phase filter before final scene auditing rather than as a standalone certificate.

A. Geometric Foundation

Let $T(q)x = R(q)x + t(q)$ denote the rigid transform induced by a configuration $q \in I$. For a reference body $C_0 \subset \mathbb{R}^3$, its pose at q is $C(q) = T(q)C_0$. We write $\mathcal{S}_I(C_0)$ for the exact swept workspace set over I and $\mathcal{C}_I(C_0)$ for its convex envelope. RBF usually works with the convex envelope because the downstream representations are AABBs, hulls, or fixed-direction slab bounds. The following construction combines two classical computational-geometry facts—capsule Minkowski decomposition and convex-hull generator bounding—in a new planning context. Both facts are used individually in swept-sphere collision libraries and in CCD work for articulated bodies [8], [9]; the novelty here is applying their combination to *C-space box validation and multi-query reuse* rather than to single-query time-step CCD.

Minkowski sum separation (radius isolation). For a capsule $C = S \oplus B_2(r)$, rigid motion commutes with the fixed Euclidean ball, giving exact two-stage sweep bounds:

$$\mathcal{S}_I(C) = \mathcal{S}_I(S) \oplus B_2(r), \quad \mathcal{C}_I(C) = \mathcal{C}_I(S) \oplus B_2(r).$$

RBF therefore bounds the zero-radius seed first and applies the link radius at the end. This identity is exact for capsules; for sharp polytopes it requires the shape to be a rounded offset body.

Convex-hull generator bound. For a convex body $P = \text{conv}(v_1, \dots, v_m)$, bounding each vertex trajectory by an outer set $\hat{\Gamma}_i(I)$ suffices to bound the swept polytope:

$$\mathcal{S}_I(P) \subseteq \text{conv}\left(\bigcup_i \hat{\Gamma}_i(I)\right).$$

For a capsule link the seed is the center segment $[a, b]$, so only two endpoint trajectories need to be bounded.

Combining these two facts yields the practical recipe. For a capsule link $[a, b] \oplus B_2(r)$, RBF bounds only the two

endpoint trajectories, forms their convex hull, and applies the radius lift:

$$\hat{\mathcal{C}}_I([a, b] \oplus B_2(r)) = \text{conv}(\hat{\Gamma}_a(I) \cup \hat{\Gamma}_b(I)) \oplus B_2(r).$$

The numerical task reduces to computing reusable envelopes for two endpoint trajectories per link, fully determined by endpoint-trajectory enclosures plus the chosen downstream representation.

Proposition 1 (Endpoint envelopes induce link interval envelopes). *Consider a rounded link $L = [a, b] \oplus B_2(r)$ and a joint interval I . If endpoint envelopes $E_a(I)$ and $E_b(I)$ satisfy $\Gamma_a(I) \subseteq E_a(I)$ and $\Gamma_b(I) \subseteq E_b(I)$, then*

$$\mathcal{S}_I(L) \subseteq \text{conv}(E_a(I) \cup E_b(I)) \oplus B_2(r).$$

Consequently, any outer representation of the right-hand side is a valid link interval envelope for the whole joint box I .

Proof. For each $q \in I$, the center segment of the moved link is the convex hull of the two moved endpoints, so $T(q)[a, b] \subseteq \text{conv}(\Gamma_a(I) \cup \Gamma_b(I))$ over the entire interval. The endpoint inclusions place this set inside $\text{conv}(E_a(I) \cup E_b(I))$. Adding the fixed radius ball commutes with the rigid-motion sweep of the capsule, giving the stated containment. $\square$

B. Endpoint Interval Envelopes

Figure 1 illustrates the AABB extraction pipeline. Endpoint interval axis-aligned bounding boxes (iAABBs) are computed by interval forward kinematics (IFK) or tighter certified AA-based variants, combined into a conservative zero-radius centerline envelope, and padded by the link radius before collision querying. The remaining numerical task is to outer-bound each endpoint trajectory over I by a reusable set $E_k(I)$. Endpoint envelopes are the common interface between kinematic propagation and the downstream link-envelope representations.

For endpoint k , let $\Gamma_k(I)$ denote its exact workspace trajectory over interval I . Any conservative set that contains $\Gamma_k(I)$ is an endpoint interval envelope. In the baseline implementation this envelope is simply an axis-aligned bounding box, because boxes are cheap to store, easy to combine, and directly reusable by all downstream link representations.

a) Interval forward kinematics (IFK).: In the current standalone API, IFK denotes the unsplit AA-FK baseline. It propagates first-order affine arithmetic along the DH chain, so correlated trigonometric terms share noise symbols and some of the wrapping error of plain interval propagation is reduced while the result remains a certified outer bound. This unsplit baseline is still the cheapest certified endpoint source, but it can loosen on wide boxes or long chains because one AA evaluation still accumulates residual over-approximation across the entire joint box.

b) Hierarchical IFK (HIFK).: HIFK further reduces wrapping by recursively bisecting the joint box and evaluating AA-FK at the leaves before hulling the child endpoint boxes. This is not the same as the unsplit IFK

baseline: a true depth-1 HIFK evaluation already splits once and hulls two leaves, whereas IFK performs one unsplit AA-FK pass on the original box. The mechanism study in section VI-C reports two representative split depths, HIFK_3 and HIFK_5, to quantify how additional hierarchical depth trades runtime for tighter certified endpoint boxes.

c) Incremental FK state reuse.: Tree descent also carries a transient FK state: interval-valued prefix transforms $[\mathbf{T}^{0:0}], \dots, [\mathbf{T}^{0:n}]$ and the per-joint DH interval matrices used to build them. The full prefix chain is computed once at the root or after a restart; when only I_k changes, prefixes before k remain valid and only the suffix is recomputed. This short-lived state is not the persistent cache itself; it only keeps endpoint materialization cheap before LECT decides whether to retain the evidence.

d) Critical-sample envelopes.: The second source estimates the endpoint box from carefully chosen boundary and interior samples. In practice these samples include interval endpoints, turning-point candidates near multiples of $\pi/2$, and a few midpoint or near-boundary probes. This often captures extrema that IFK wraps away, at modest extra cost. The limitation is completeness: separable per-joint samples do not certify coupled extrema. CritSample is therefore a performance-oriented source whose reported planning paths are discharged by the final scene audit unless an external certificate is added.

e) Analytical extrema.: The third source solves endpoint-coordinate extrema from the trigonometric structure of the chain. Because each revolute-joint transform is affine in $(\cos q_j, \sin q_j)$, an endpoint coordinate becomes a simple sinusoid when one joint is free and a low-degree bilinear trigonometric polynomial when two joints remain free. This yields a structured solver: vertices and 1D boundary edges can be enumerated exactly, 2D faces reduced to polynomial root finding, and higher-dimensional residual cases handled by local descent with interval verification or branch-and-bound closure. Analytical bounds are the tightest of the three sources, but also the most expensive, so they are better suited to offline certification and profiling than to the online path.

Taken together, these sources define a clear spectrum: IFK is the cheapest certified AA baseline, HIFK_3 and HIFK_5 offer progressively tighter certified outer bounds at increasing cost, CritSample is the cheapest tighter advisory bound, and the analytical source is the tightest but most expensive. Endpoint envelopes are the right intermediate abstraction: they are the state variable of the joint-box-to-envelope map, so the downstream link-envelope layer stays fixed while the upstream source changes with the certification and runtime budget.

C. Link Interval Envelopes

Once endpoint envelopes are available, the remaining design question is how to represent the swept link volume compactly for box growth, collision filtering, and optional reuse. For a capsule link, RBF does not reason about

the thick link directly. It first bounds the zero-radius center segment between the two endpoint envelopes and then pads that centerline envelope by the link radius. A conservative link interval envelope $B_\ell(I)$ is any set that contains the full swept capsule over interval I . In the default box-based implementation this is written as $B_\ell(I) = B_\ell^\circ(I) \oplus \mathcal{R}(r_\ell)$, where $B_\ell^\circ(I)$ bounds the centerline sweep and the padding operator $\mathcal{R}$ becomes B_∞ for AABB envelopes.

The following representations share the same endpoint evidence but preserve different amounts of downstream geometric detail.

a) AABB envelope.: AABB envelopes are the simplest option. The ordinary LinkAABB construction takes the two endpoint envelopes, wraps them in one box, and pads that box by the link radius:

$$B_\ell(I) = \text{bbox}(E_{\ell-1}(I) \cup E_\ell(I)) \oplus B_\infty(r_\ell).$$

This representation can be tightened by subdividing the reference center segment into S pieces, bounding each piece separately, and retaining the union of the padded sub-envelopes. Larger S preserves more directional variation along the link while keeping reconstruction cheap once endpoint envelopes are available. The main caveat is that the subdivision must remain explicit: if all sub-envelopes are collapsed back into one global AABB, much of the gain disappears.

b) KDOP slab envelope.: A k -discrete-oriented-polytope (k -DOP) envelope stores lower and upper support values along a fixed set of directions. Here the 26-plane variant (KDOP26) uses the three coordinate axes, six face-diagonal axes, and four body-diagonal axes, for a total of 13 unsigned axes and 26 supporting planes. Endpoint boxes are projected onto these axes, padded by the link radius, and merged over each retained subsegment. Collision checking is correspondingly cheap: after the ordinary link AABB test, the slab test continues with the non-axis directions and exits on the first separating interval. The staged order first tests the DOP18 axes and only reaches the final DOP26 diagonals for near survivors.

c) SupportHull envelope.: When the endpoint envelopes are convex, their convex hull preserves directional structure that a single AABB discards. SupportHull is the compact runtime realization of that idea: instead of storing an explicit dense hull, it keeps the two endpoint AABBs and the link radius as a support-function record. For a query direction d , the support point is the better of the proximal and distal AABB support points, and collision refinement against an obstacle AABB uses a fixed-size Gilbert–Johnson–Keerthi (GJK) narrow phase with the obstacle inflated by the link radius and safety margin. This retains much of the convex-hull directional fidelity while keeping the default record compact; a hybrid variant may additionally retain KDOP26 intervals as a mid-phase before GJK, but that is treated as a separate runtime/storage point rather than as the default SupportHull envelope.

In this way, the link-envelope layer is separate from the upstream kinematics solver. Sections III-A to III-C define a modular endpoint-to-link envelope map from joint-space boxes to reusable workspace filters. Once endpoint envelopes are available, the remaining design choice is the representation and the corresponding planning-time cost/tightness trade-off.

IV. Lifelong Envelope Cache Tree

The planner can materialize the envelopes of section III online. LECT is a secondary optimization for workloads in which repeated builds revisit similar joint boxes or use envelope representations expensive enough to replay. It stores scene-independent endpoint/link-envelope evidence keyed by robot kinematics and joint intervals. Scene-specific obstacle tests, box labels, graph connectivity, and final-audit outcomes remain outside the cache. This section therefore records only the cache interface used by the planner; the scientific claim of the paper does not depend on LECT providing broad cross-scene transfer.

A. Cache Key and Split Policy

LECT is a dynamically grown binary KD tree over the joint-limit box $\mathcal{Q}$. Each node represents one joint interval and stores interval bounds, split metadata, cached endpoint evidence from which link envelopes can be reconstructed, and residency state used by the implementation. The useful property is that the stored evidence is *kinematics-keyed* rather than *scene-keyed*: later scenes may replay a materialized envelope record, but they must rerun their own scene-local obstacle tests.

The split policy is envelope-aware rather than purely geometric. For a parent interval I , let I_d^- and I_d^+ be the two children obtained by splitting joint dimension d at its midpoint, and let $\mathcal{V}(I) := \sum_{\ell} \text{vol}(B_{\ell}(I))$ be the cumulative downstream envelope volume. The default best-tighten score chooses the split with the smallest worst-child envelope burden,

$$d^* = \arg \min_d \max \left\{ \sum_{\ell} \text{vol}(B_{\ell}(I_d^-)), \sum_{\ell} \text{vol}(B_{\ell}(I_d^+)) \right\}.$$

The implementation uses this score in a depth-synchronous schedule so that all nodes at the same depth share one split decision. Candidate dimensions are also filtered by FK reachability: splitting a joint beyond the last active frame that contributes to any retained link endpoint cannot tighten the endpoint envelopes and is skipped.

B. Runtime Use and Validation Boundary

LECT is queried by FINDFREEBOX, not by the graph search itself. Given a joint interval, the cache either replays endpoint evidence or materializes it online, then derives the requested link representation. Directional records are used as refinement evidence rather than as standalone free-space certificates: the runtime first evaluates padded link AABBs, then uses KDOP26 slabs or SupportHull/GJK

only for survivors of the AABB test. When a node is split, child envelopes may refine the parent witness, but any compact parent record remains representation-dependent and is rechecked against the current obstacle set before a box can enter the forest.

C. Storage and Warm-Build Semantics

Persistence is selective. Endpoint evidence is the common reusable record, but simple IFK+AABB evidence can be cheaper to recompute online than to reload. KDOP26 and SupportHull records are retained only when their tighter filters are expected to amortize storage and lookup cost. Implementation details such as node arrays, residency management, and optional symmetry compression are artifact-level engineering choices; they are not used as independent evidence for the paper's planning claim.

The prewarm phase follows the same separation. It materializes FK/link-envelope evidence only on the target leaf layer and reconstructs upper layers by child-hull propagation during checkpoint. This avoids charging direct FK at multiple depths for the same cache and keeps parent witnesses tied to the leaf records that generated them.

In the warm-build path, LECT reuses only kinematic envelope evidence. On a cache hit, the current link-envelope representation is reconstructed from stored endpoint records and then revalidated against the current obstacle set. A warm build still regrows and revalidates scene-specific RBF boxes and never treats a persisted box label as a free-space certificate. In section V, LECT serves only as the materialization layer; obstacle checking, occupancy metadata, adjacency bookkeeping, and regrowth reactions to changing $\mathcal{O}$ remain inside the box-region forest routines. For the reviewer-sensitive Shelf+IIWA protocol, the external d18 cache is treated as read-only during evaluation: warm shelf rows disable active-database backfill and therefore cannot silently accumulate new online envelope evidence while being measured. The balanced random-scene study keeps that writeback path enabled because it measures cache growth on novel scenes rather than a fixed held-out shelf target.

V. RapidBoxForest Construction

This section describes the planning layer of RBF. The algorithm grows a scene-specific forest of C -space boxes and exports the forest as a graph for repeated corridor queries. The growth loop is RRT-like, but the state added to the data structure is a free region rather than a point or edge: samples select frontiers, the link-envelope oracle materializes a usable box, and accepted boxes extend a connected region graph. In adaptive cell-decomposition language, the boxes are cells, FindFreeBox is the local refinement and classification primitive, and the forest graph is the adjacency structure used for query search. The link interval envelopes of section III and the optional LECT cache of section IV serve this planner as collision-filtering and materialization tools.

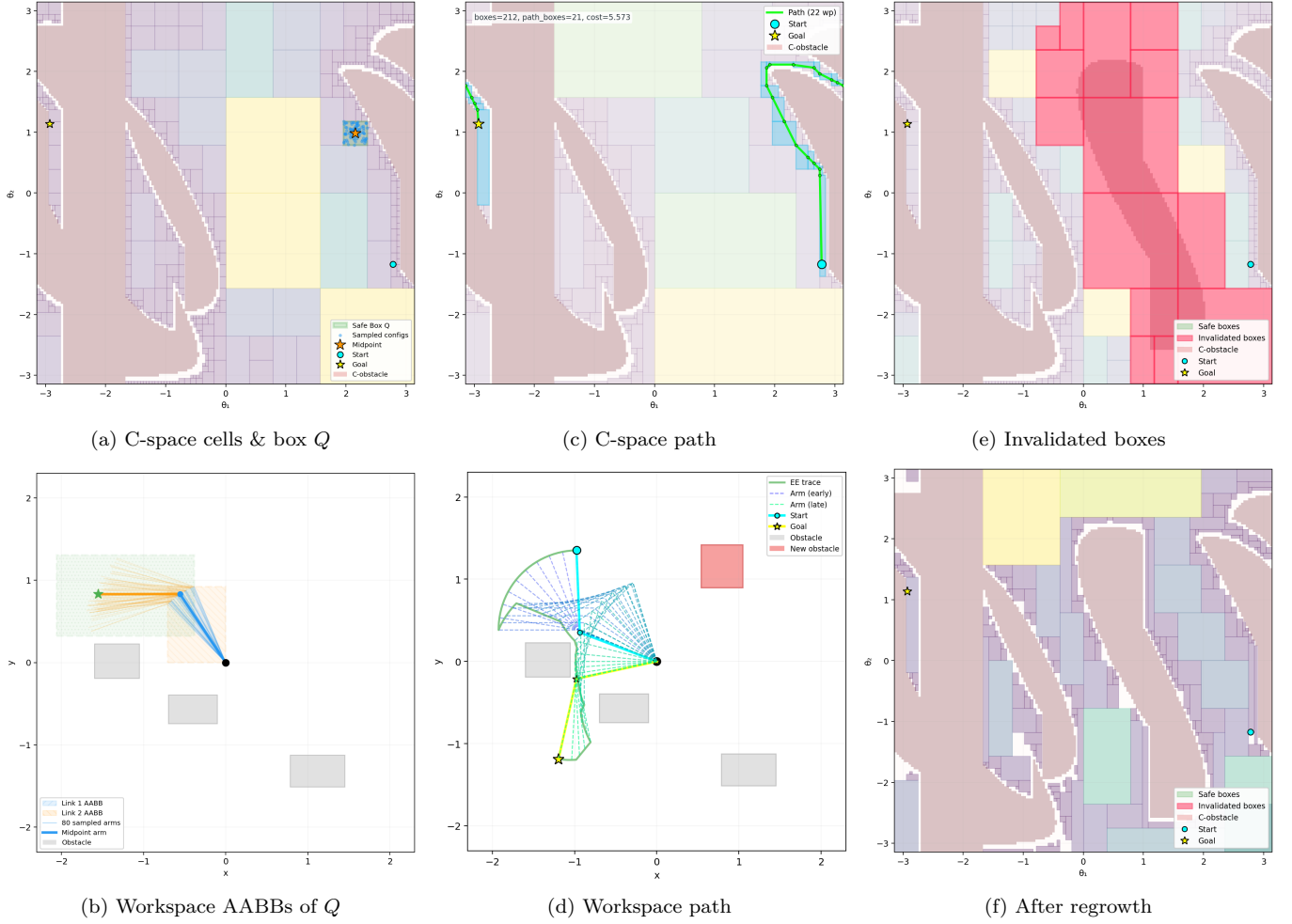


Fig. 2. Illustrative RBF build, query, and obstacle-update cycle on a planar 2-link manipulator. (a, b) Adaptive C -space cells and workspace AAB envelopes of the highlighted box Q . (c, d) Box-corridor path in C -space and its workspace trajectory. (e, f) After obstacle insertion, intersecting boxes are invalidated (red), and localized repair/regrowth updates the explicit forest without requiring a full rebuild.

Throughout this section, a validated box means a box accepted by the selected scene-validation policy. In strict conservative mode, paths that remain inside the validated box union inherit the corridor statement below. In all reported planning experiments, paths that leave this union through repair, smoothing, or external composition are charged where used and passed through the same final scene audit.

Conceptually, the build process has three stages. First, a seed-guided local oracle adaptively refines and classifies a C -space cell around a selected configuration. Second, repeated frontier expansion grows these cells into a connected forest, giving the method its region-valued RRT/PRM character. Third, consolidation coarsens redundant boxes before the structure is exported as a query graph. The consolidation pass is part of the reusable build phase and its time is included in reported build measurements.

A. Construction Objective

Let $\mathcal{Q}$ denote the joint-limit box and let $\mathcal{O}$ denote the current obstacle set. The construction objective is to build

a finite family

$$\mathcal{F} = \{B_1, \dots, B_N\}, \quad B_i \subseteq \mathcal{Q},$$

such that every B_i is a validated free-space box, the union $\bigcup_i B_i$ covers as much useful query-relevant free space as possible under the available box, time, and depth budgets, and the induced adjacency graph supports efficient downstream search. This object is scene-specific: changing $\mathcal{O}$ changes which intervals are accepted into the forest, even though the underlying envelope evidence in LECT remains reusable.

The central structural rule is that the forest is not merely a free-space cover, but a *corridor-compatible* free-space cover. Boxes are inserted under a connected-box constraint.

Viewed as adaptive cell decomposition, this objective is intentionally workload-biased rather than globally exhaustive. The planner does not try to classify every cell in $\mathcal{Q}$. It refines and accepts cells near useful frontiers, task seeds, and connector candidates, then keeps the resulting adjacency graph as the reusable planning object.

a) *Connected-box rule.*: For boxes

$$B_i = \prod_{d=1}^n [l_i^{(d)}, u_i^{(d)}], \quad B_j = \prod_{d=1}^n [l_j^{(d)}, u_j^{(d)}],$$

Algorithm 1 FindFreeBox: local box materialization around a seed.

Input: Seed configuration q , LECT T , obstacle set $\mathcal{O}$, depth cap $D_{\max}$

Output: Validated interval I_v or failure

```

1:  $v \leftarrow \text{root}(T)$ ;  $d \leftarrow 0$ ; initialize current interval and FK state
2: while  $d \leq D_{\max}$  do
3:   Cache materialization
4:   materialize the endpoint and link envelopes of  $v$  if needed
5:   Scene validation
6:   if  $v$  is marked fully occupied then
7:     return failure
8:   end if
9:   if  $v$  is validated collision-free with respect to  $\mathcal{O}$  then
10:    return interval  $I_v$ 
11:   end if
12:   Refinement
13:   if  $d = D_{\max}$  then
14:     return failure
15:   end if
16:   if  $v$  has no children then
17:     split  $v$  along the depth-synchronous split dimension of depth  $d$ 
18:   end if
19:    $k \leftarrow$  split dimension of  $v$ 
20:    $v \leftarrow$  the child of  $v$  whose interval contains  $q$ ; update the current interval
21:   update the FK state from dimension  $k$  onward;  $d \leftarrow d + 1$ 
22: end while
23: return failure

```

RBF treats them as connected when, in every coordinate, the two intervals either overlap or are separated by at most a small tolerance ε_c :

$$\max\{l_i^{(d)}, l_j^{(d)}\} \leq \min\{u_i^{(d)}, u_j^{(d)}\} + \varepsilon_c, \quad d = 1, \dots, n.$$

When $\varepsilon_c = 0$, this reduces to ordinary interval intersection or boundary touching in every dimension. A small positive tolerance absorbs the boundary offset used by grower seeds and floating-point roundoff.

This rule is deliberately weaker than requiring a full shared face. It treats overlap, touching, and tolerance-scale gaps uniformly as graph-connectivity evidence, while validity remains attached to the accepted box union itself. The output of the construction stage is a forest of validated boxes whose graph connectivity is aligned with the corridor object queried later, without imposing a stricter face-sharing invariant than the grower actually maintains.

B. Find Free Box

FINDFREEBOX descends LECT from the root toward the leaf containing seed $q \in \mathcal{Q}$, materializing endpoint and link envelopes lazily along the path, skipping saturated subtrees via occupancy metadata, and returning the current interval as a validated box candidate as soon as the scene oracle accepts it. Tree topology and endpoint records are reused from the scene-independent cache, while scene-specific validation is deferred to each grower visit. Algorithm 1 gives the complete pseudocode.

The termination behavior is straightforward. Under depth cap $D_{\max}$, each loop iteration either returns immediately or descends one more level in the tree, so the procedure can make at most $D_{\max} + 1$ decisions. In strict conservative mode, any returned interval has passed envelope disjointness against $\mathcal{O}$; in performance-oriented

modes, the returned interval is a validated planning region that remains subject to the common final audit used for reported paths.

C. Forest Growth

Starting from an initial seed set, the forest is grown by repeatedly invoking the find-free-box procedure on carefully chosen candidate configurations. Each seed first produces a root box through the find-free-box procedure, and these root boxes initialize one or more connected components of the forest. The main grower should be understood as an evidence-guided rapidly exploring process rather than as a generic random cover: every iteration chooses a frontier component, proposes a nearby interval worth testing, and accepts the result only when it extends the forest by a usable connected corridor segment.

Each iteration makes two explicit heuristic decisions: it samples a target configuration q_t , then chooses a boundary seed on an existing box face that is closest to that target. The reported planning configuration uses a sequential Bernoulli schedule with component-connector probability $p_c = 0.45$, query-root bias $p_g = 0.20$, intertree-root bias $p_i = 0.25$, and unexplored-volume probability $p_u = 0.45$. A component-connector target between disconnected root components is attempted first. If no connector target is used, coordinated multi-tree mode may target a root from another active component; because $p_i \leq 0.5$ in the reported runs, no high-bias pulse is active and the sustained cap is also 0.25. If no intertree target is used, the grower then tries a query-root target, then a LECT unexplored-volume leaf, and finally a uniform root-interval sample. The same reported configuration uses 5000 boxes, a 60 s build timeout, an FFB depth cap of 120, eight worker threads, a component-connector candidate limit of four, a staged component-connect radius of 0.35 in normalized L_∞ distance, a post-connect time budget of 450 ms, and a quality floor of 64 connected boxes. These numbers are planner settings rather than envelope-theory parameters, but they are part of the self-contained specification for the reported rows.

Given a target q_t and an existing box $B = \prod_j [\ell_j, u_j]$, the grower enumerates feasible faces $f = (B, d, s)$, where d is a joint dimension and $s \in \{-1, +1\}$ is the lower or upper side. Let τ be the adjacency tolerance and $\epsilon_b = \max(\epsilon_{\text{guard}}, \tau/4)$. A face is feasible only if stepping outside it by ϵ_b remains inside the global root interval. In the reported configurations $\epsilon_{\text{guard}} = 0$, so this formula uses the $\tau/4$ floor. The seed associated with the face is

$$q_f[j] = \begin{cases} u_d + \epsilon_b, & j = d, s = +1, \\ \ell_d - \epsilon_b, & j = d, s = -1, \\ \text{clip}(q_t[j], \ell_j + \epsilon_b, u_j - \epsilon_b), & j \neq d, \end{cases} \quad (1)$$

with clipping to $[\ell_j, u_j]$ when a box is thinner than $2\epsilon_b$. Faces are ranked by the squared target distance

$$\sigma(f; q_t) = \|q_t - q_f\|_2^2, \quad (2)$$

Algorithm 2 GrowForest: budgeted expansion of a connected box-region forest.

Input: LECT T , obstacle set $\mathcal{O}$, seed set $\mathcal{S}$, box budget $N_{\max}$, time budget $t_{\max}$
Output: Validated box-region forest $\mathcal{F}$

```

1:  $\mathcal{F} \leftarrow \emptyset$ ; initialize connected components from  $\mathcal{S}$ 
2: for all  $q \in \mathcal{S}$  do
3:    $B \leftarrow \text{FindFreeBox}(q, T, \mathcal{O})$ 
4:   if  $B \neq \emptyset$  then
5:     insert  $B$  into  $\mathcal{F}$  as a root box
6:   end if
7: end for
8: while  $|\mathcal{F}| < N_{\max}$  and time budget not exhausted do
9:   draw target  $q_t$ : connector with prob.  $p_c$ , query/intertree root with prob.  $p_g$  or  $p_i$ , unexplored LECT leaf with prob.  $p_u$ , otherwise uniform
10:  enumerate feasible faces  $f = (B, d, s)$  of frontier boxes and compute  $q_f$  and  $\sigma(f; q_t)$ 
11:  choose the first of the 128 lowest-score face seeds that is not cooled and not in the frontier-seed cache
12:  set  $(q_{\text{seed}}, B_{\text{near}}) \leftarrow (q_f, B)$  for that face
13:   $B_{\text{new}} \leftarrow \text{FindFreeBox}(q_{\text{seed}}, T, \mathcal{O})$ 
14:  if  $B_{\text{new}} \neq \emptyset$  and  $\text{BoxesConnected}(B_{\text{new}}, B_{\text{near}}, \tau)$  then
15:    insert  $B_{\text{new}}$  into  $\mathcal{F}$ ;  $\text{UpdateComponents}(\mathcal{F})$ 
16:  else
17:    record the failed LECT leaf; cool it after one failed attempt for a 300-box horizon
18:  end if
19: end while
20: return  $\mathcal{F}$ 

```

and the implementation retains the 128 lowest-score face candidates across the scanned frontier boxes. It then tries them in increasing score order, skipping a candidate if the corresponding LECT leaf is temporarily cooled or if a component-connection cache has already recorded the same boundary seed. The first remaining candidate becomes the seed for FINDFREEBOX.

The candidate box is accepted only if FINDFREEBOX validates it and the new box is connected to its parent under the same adjacency predicate used by the query graph: all dimensions overlap up to tolerance τ , and at least one dimension touches within tolerance or all dimensions overlap. Failures are also handled explicitly. When the LECT leaf containing a failed seed reaches the hard-frontier failure threshold of one failed attempt, the leaf is skipped for a horizon of 300 subsequently accepted boxes, unless a later depth stage raises the active search depth and enables a retry. Cooling and the frontier-seed cache do not alter the validation boundary; they only reduce repeated calls on already failed or already covered frontier regions. The resulting method is summarized in algorithm 2.

The main reported configuration uses coordinated multi-goal growth. Five seed configurations initialize the shelf roots, and the grower uses the fixed resource budget reported in section VI-A and fig. 3. This choice keeps the growth schedule simple enough to preserve one authoritative forest state while still exposing reusable LECT evidence across threads and cache-warm reruns. Staged wavefront growth and task-batched multi-threaded growth are treated as engineering variants, and the parallel-scaling experiment reports the present limit of the task-batched variant. The output of this section is the query-ready box-region object only.

Algorithm 3 ConsolidateForest: graph-preserving promotion, merging, and pruning.

Input: Validated forest $\mathcal{F}$, LECT T , obstacle set $\mathcal{O}$
Output: Consolidated validated forest $\mathcal{F}'$

```

1:  $\mathcal{F}' \leftarrow \mathcal{F}$ 
2: Stage 1: sibling promotion
3: repeat
4:   promote any validated sibling leaves whose parent is collision-free
5: until no eligible sibling pair remains
6: Stage 2: validated connected/contact merging
7: merge exact connected pairs whose union is itself a single box
8: for all connected candidate pairs  $(B_i, B_j)$  in  $\mathcal{F}'$  do
9:   build a candidate parent region  $H$  and score its volume expansion
10:  if the chosen envelope representation of  $H$  is collision-free with respect to  $\mathcal{O}$  then
11:    replace  $B_i, B_j$  by  $H$ 
12:  end if
13: end for
14: Stage 3: containment pruning
15: remove only contained boxes whose adjacency and segment witnesses transfer to retained covering regions
16: return  $\mathcal{F}'$ 

```

D. Forest Consolidation

Raw growth tends to produce a forest that is finer than necessary. The purpose of consolidation is to replace a large collection of locally valid boxes by a smaller collection of equally valid but more useful regions. Two operations are central.

The first is *promotion*. When two occupied sibling leaves are both collision-free and their parent interval is also collision-free, the two leaves can be replaced by the parent. This is the most local form of coarsening and directly reuses the multiresolution structure of LECT.

The second is *multi-level coarsening*. Exact connected merges are the cleanest case: if two boxes are connected and their union is itself one box, the merged region can be accepted directly after validation. More general connected candidate merges are handled by constructing a candidate parent region and accepting it only when its chosen envelope representation remains collision-free against the current obstacle set. This is where the representation choices of Sections III and IV become operational. In an AABB-based forest, the compact parent witness is typically the enclosing AABB of the two child boxes. KDOP parents can merge child slab intervals by per-axis minima and maxima, while SupportHull parents are kept compact only when the merged endpoint-hull witness remains acceptable; otherwise the child witnesses are retained. Greedy non-sibling candidates are ranked by a volume-expansion score, so exact face merges are preferred and high-inflation parents are deferred unless they pass validation under the scene oracle. The consolidation logic is summarized in Algorithm 3. The final containment-pruning step removes only boxes that are fully covered by accepted validated regions and whose graph incidence can be transferred to a retained covering region. It changes neither the represented free-space union nor the corridor witnesses used by query search.

Validity follows directly: every accepted promotion or merge has itself passed the scene oracle, so the represented

free-space union cannot shrink. Containment pruning removes only boxes fully covered by a retained region that inherits their adjacency witnesses, preserving all corridor connectivity.

E. Corridor Query

After growth and consolidation, the forest is exported as a graph $G = (V, E)$ with one vertex per validated box. Two vertices are connected by an edge when their boxes satisfy the connected-box rule above. Lightweight querying is possible because this connected-box structure is enforced during growth and consolidation rather than repaired after graph export.

If disconnected adjacency islands remain after growth, an optional connector stage attempts targeted bidirectional sampling bridges between nearest frontier boxes of different components. A successful bridge is stored as a segment edge rather than as a claim that the two boxes overlap. This distinction is important for both the RBF query layer and the GCS negative controls: a segment edge is a collision-checked path witness between two boxes, whereas a GCS edge over convex regions requires feasible overlap of the regions themselves.

This pair $(\mathcal{F}, G)$ is the final planning object produced by the build stage. Querying it is intentionally lightweight. Given start and goal configurations q_s, q_g , the planner associates them with start-side and goal-side boxes in $\mathcal{F}$ and then searches G for a connecting box sequence. The query problem is reduced to box-corridor retrieval inside an already constructed free-space representation.

A shortest path on G yields a box sequence whose union defines a corridor between the two query endpoints. A geometric path is then recovered as a waypoint chain inside that corridor. In the experiments, this candidate is then passed through the same OMPL-style simplification/local post-processing and fixed-step final audit used by the comparison pipelines. That post-processing can move the final reported path outside the source box union. Consequently, the box-corridor statement below applies only to contained paths before or after post-processing, while the reported planning success in the main curves is the common fixed-resolution audit outcome.

Theorem 1 (Conditional validated box-corridor path). *Let $\mathcal{F}$ be a forest whose boxes have each passed conservative link-envelope validation against obstacle set $\mathcal{O}$. If a continuous path $\gamma : [0, 1] \rightarrow \mathcal{Q}$ satisfies $\gamma([0, 1]) \subseteq \bigcup_{B \in \mathcal{F}} B$, then every configuration on γ is collision-free with respect to $\mathcal{O}$.*

Proof. For any $s \in [0, 1]$, the containment assumption gives a validated box $B \in \mathcal{F}$ with $\gamma(s) \in B$. By construction, validation of B means every conservative link envelope for B is disjoint from $\mathcal{O}$. Because each such envelope contains the link geometry for every configuration inside B (section III), the robot geometry at $\gamma(s)$ is also disjoint from $\mathcal{O}$. Since s was arbitrary, the whole path is collision-free. $\square$

a) *Reporting protocol.*: Theorem 1 is a conditional statement: it covers path portions contained in a conservatively validated box union. It is not the acceptance rule for the main experimental comparisons. When OMPL simplification, local repair, smoothing, or external composition leaves that union, the path is accepted only if it passes the same fixed-resolution final scene collision audit used for the baselines. The main performance curves should therefore be read as post-processed, fixed-audit planning results rather than as evidence that every reported path satisfies the box-corridor containment hypothesis.

For dynamic scenes, the same explicit forest representation supports local graph maintenance without compulsory rebuilds. Each accepted box records the obstacle evidence that blocked or validated its envelope tests; obstacle deletions cannot invalidate existing boxes accepted under the same validation policy, because free space only grows. Deletion triggers only a spatial dirty-region pass around the removed obstacle to choose local regrowth seeds, after which any new box is validated by the same oracle as in the original build. Obstacle insertion is stricter: the new obstacle is inflated by the dirty-region padding, all overlapping boxes are rechecked exactly, invalid boxes and incident graph edges are removed, and every surviving segment edge is re-audited in the edited scene. The repair stage then regrows from deleted-box centers, live neighbours, original task seeds, and component-boundary candidates; each committed box must pass the active validation policy before it enters the forest. Finally, query failure can trigger a bounded local repair search, again filtered by the same final audit. Warm rebuild remains available as a measured baseline and fallback policy.

If no contained conservative corridor is recovered, no box-corridor guarantee attaches to that query. The construction still exports a concrete box-region planning object, and section VI evaluates its build cost, query cost, path quality, and success rate after the same post-processing and fixed-resolution final audit used for the other planners.

VI. Experiments

This section evaluates RBF as a multi-query planning system whose returned paths are compared after common post-processing and fixed-resolution final audit. The primary question is the design point: how much reusable build/query time can the implemented RBF pipeline save on the tested workloads, and what audited path-length gap does it pay? The endpoint/link envelope, LECT, and dynamic-update studies explain construction cost and maintenance behavior within the same scoped system evaluation.

a) *Scope.*: The main evidence consists of the Shelf+IIWA and IIWA/UR5/Panda anytime trade-off figures with accompanying compact summary tables. Mechanism results report endpoint interval-AABB behavior, the CritSample-based envelope representation trade-off, incremental IIWA LECT reuse, and a dynamic-update capability comparison. Together these experiments

support a controlled workload claim about planning throughput, audited path quality, repeated-query amortization, and mechanism-level costs. They do not establish distributional generalization or sensitivity to random seed count, audit resolution, worker count, row-selection rule, grower strategy, or narrow clearance.

b) Reporting Protocol.: The paper uses two reported success notions and one theorem boundary. *Candidate success* means that a planner produced a candidate path under its own construction rules; it is a search/solver-completion statistic and is not credited as safe motion. *Final-audited success* is the common reported planning metric for all methods: after the method-specific construction and the shared OMPL-style simplification/local post-processing, every returned path is checked by the same scene collision oracle at a fixed 0.01 joint-space segment step. This is a uniform empirical safety check at the stated resolution, not a continuous certificate. The conservative box-corridor theorem is a narrower boundary for RBF paths that both use conservative validation and remain inside the validated box union. It is not used as the acceptance criterion for the main curves, because the common post-processing, bridge repair, smoothing, and external composition can leave that union. The paper-wide audit accounting accepts 663/668 counted paths. The five counted failures come from the direct merger+GCS negative-control rows in the accounting suite and are not credited as planning successes; unsafe solved GCS attempts that are rejected before fallback are tracked separately. A closure scan of the current artifacts found no stored corridor-containment fields, so the main planning tables report fixed-resolution final-audited rows only and do not measure the fraction of those rows that would also stay inside the validated box union. We also re-audited 3240 stored random-scene checkpoint paths at several segment steps. The stated 0.01 audit step produced 0 invalid records. Finer checks exposed resolution sensitivity shared by the pipelines: 0.005 and 0.002 steps found 502 and 678 invalid checkpoint records, respectively, corresponding to 79 and 111 unique stored paths. Thus the reported success statistics should be read exactly as fixed-resolution final-audit statistics, not as continuous safety guarantees or guarantees that the post-processed path remains inside the box union.

c) Platform and Scenes.: All experiments reported here run on an Intel Core i5-12600KF workstation with 16 logical CPUs and 31 GiB RAM. Unless the study itself is the parallel-scaling sweep, all planners use eight worker threads. Shelf+IIWA uses one canonical scene with five task pairs and manually selected reference anchors from the Marcucci scene. The random-scene study spans IIWA/UR5/Panda easy, medium, and hard settings with 50 fixed random scene seeds per setting, where easy/medium/hard denote incrementally generated 4/8/12-obstacle scenes. These counts support a controlled system comparison, but they do not by themselves establish broad distribution-level generalization.

d) Baseline Accounting and Seed Policy.: IRIS-NP+GCS uses the Drake implementation and charges region build. PRM, RRTConnect, and BIT* use OMPL. We separate reusable multi-query pipelines (RBF, PRM, IRIS-NP+GCS) from no-build single-query references (RRTConnect and BIT*): PRM charges roadmap build, IRIS-NP+GCS charges convex-region build, and RBF charges box-forest build, while RRTConnect and BIT* are treated as single-query references. RRTConnect uses one maximum-timeout call per seed/query with no restart accounting and stops on first connection; a timeout is a failure. BIT* uses one fixed-timeout call per seed/query, and the plotted curve/table rows are taken from the audited monotone incumbent checkpoint trace rather than raw checkpoint stage values. The IRIS comparison should be read as a comparison of the implemented pipelines under a shared workload-level seed protocol. In the Shelf+IIWA workload, all planners use the same task pairs and the same manually selected Marcucci reference anchors, with query-derived midpoint seeds when a planner requires additional initial points. In the random-scene workload, all planners use the same fixed random scene/query seeds. This common seed protocol improves baseline fairness, but the build-time gaps remain full-pipeline results because the region representations, graph construction, repair, and solver implementations differ.

e) Acceptance Boundary.: The main comparison accepts paths by the same fixed-resolution final audit for all methods. Paths inside conservatively validated RBF corridors inherit the box-corridor statement, but the shared post-processing step can move a returned path outside that union. Such paths have only the common final-audit semantics, so the theorem is not transferred to the main curves.

f) Tabulated Rows and Omitted Sensitivities.: Where a table accompanies an anytime figure, the table gives a compact numeric slice of the corresponding curve rather than a separate evaluation. The tabulated-row extraction was postprocessed over 820 anytime stage points. The reported rows were unchanged when the path-tolerance rule was varied over 5%, 8%, 10%, and 15%; a zero-tolerance shortest-path-only rule changed 25/50 scenario-method rows. At the reported rows, the RBF-SH path gap to the shortest tabulated method has median 0.056 rad, 90th percentile 0.296 rad, and maximum 0.455 rad on Shelf+IIWA. These checks support the use of the tables as compact numeric slices, while the full curves remain the primary evidence. The present paper still does not exhaustively characterize sensitivity to seed count, worker count, grower strategy, narrow clearance, or broader audit rules.

A. Shelf+IIWA Efficiency and Path Quality

Figure 3 compares Shelf+IIWA anytime incumbents on the canonical Marcucci workload and shows the corresponding amortization view. This figure is the first main

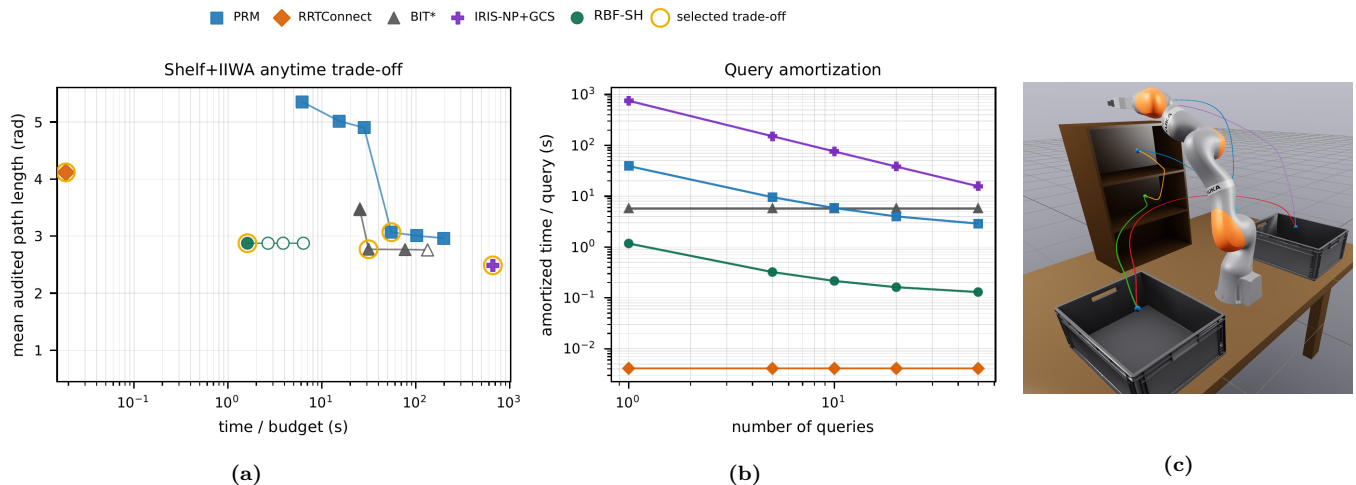


Fig. 3. Shelf+IIWA final-audited design-point trade-off. (a) Anytime trade-off under the method-specific build/query protocol; points nearer the lower-left corner combine lower charged time with shorter audited path length. Filled markers indicate incumbent-improving checkpoints; open markers show a sparse representative subset of nonimproving checkpoints. (b) Amortized per-query cost for the tabulated rows. (c) Canonical Marcucci scene used for the Shelf+IIWA experiments. Gold rings mark the rows reported in table I. AS, TS, CS, LB, and RB denote the above-shelf target, top shelf, middle shelf, left bin, and right bin, respectively.

TABLE I

Common-rule tabulated Shelf+IIWA rows from Fig. 3, reported by query. Gold rings in Fig. 4 mark these rows only to expose detailed numeric values; the full curves remain the primary evidence. RRTConnect uses one max-timeout attempt per seed/query and returns on connection; BIT* uses one fixed-timeout invocation per seed/query with an audited monotone incumbent checkpoint trace. OMPL planning, final simplify, and fixed-resolution final audit use the same 0.01 joint-space segment step. Query and path entries are success-only averages over paths that pass this fixed-resolution audit.

	SBF-SH		IRIS-NP+GCS		PRM		RRTConnect		BIT*	
Query	Query (s)	Path	Query (s)	Path	Query (s)	Path	Query (s)	Path	Query (s)	Path
AS→TS	—	—	—	—	—	—	—	—	—	—
TS→CS	—	—	—	—	—	—	—	—	—	—
CS→LB	—	—	—	—	—	—	—	—	—	—
LB→RB	—	—	—	—	—	—	—	—	—	—
RB→AS	—	—	—	—	—	—	—	—	—	—

planning result: it tests whether the box-region forest reaches a low charged-time design point and what path-length premium it pays. All methods in this figure are compared after the common final-path post-processing and fixed 0.01 joint-space audit protocol; OMPL planners use the same segment step for planning and final simplify. RRTConnect uses one maximum-timeout call for each seed/query and exits as soon as a connection is found. BIT* uses one fixed-timeout call per seed/query, and the plotted line is the audited monotone incumbent summary across checkpoints rather than raw checkpoint-stage values. Filled markers denote incumbent-improving checkpoints, while open markers show only a sparse representative subset of nonimproving checkpoints. Panel (a) shows the anytime trade-off, panel (b) reports amortized per-query cost for the tabulated rows, and table I gives the corresponding success-only averages over final-audited trials. In the tabulated Shelf+IIWA slice, RBF-SH uses a 1.07 s reusable build and sub-0.102 s online queries, but its audited paths are longer than the IRIS-NP+GCS paths for all five listed queries and are not uniformly shorter than the BIT* incumbents. The Shelf result should therefore be read as a latency/amortization trade-off, not as a shortest-path claim.

B. Balanced Random-Scene Pipeline Study

Figure 4 reports balanced easy/medium/hard IIWA, UR5, and Panda scenes on the same platform as Shelf+IIWA. The random-scene comparison again uses the common final-path post-processing and fixed 0.01 joint-space audit protocol; OMPL planners use the same segment step for planning and final simplify. RRTConnect is one maximum-timeout call per scene/trial with no restart accounting, while BIT* is one fixed-timeout call with incumbent checkpoints summarized as a monotone anytime curve. For table II, each method and scenario contributes one row chosen by a common rule: among points within 8% of the shortest audited path, select the minimum of a single path/log-time utility.

The easy/medium/hard labels control obstacle count and nominal clutter, not a guaranteed runtime ordering. In easy scenes the larger free space can leave sampling-based planners and region builders with more irrelevant volume to explore before the useful corridor is resolved. As a result, an easy panel can charge more build or search time than the corresponding medium or hard panel even under the same planner settings.

Under this shared tabulation rule, table II summarizes the reported trade-offs. In this pipeline-level comparison, the RBF rows build in 0.147–0.509 s and query in 0.0061–

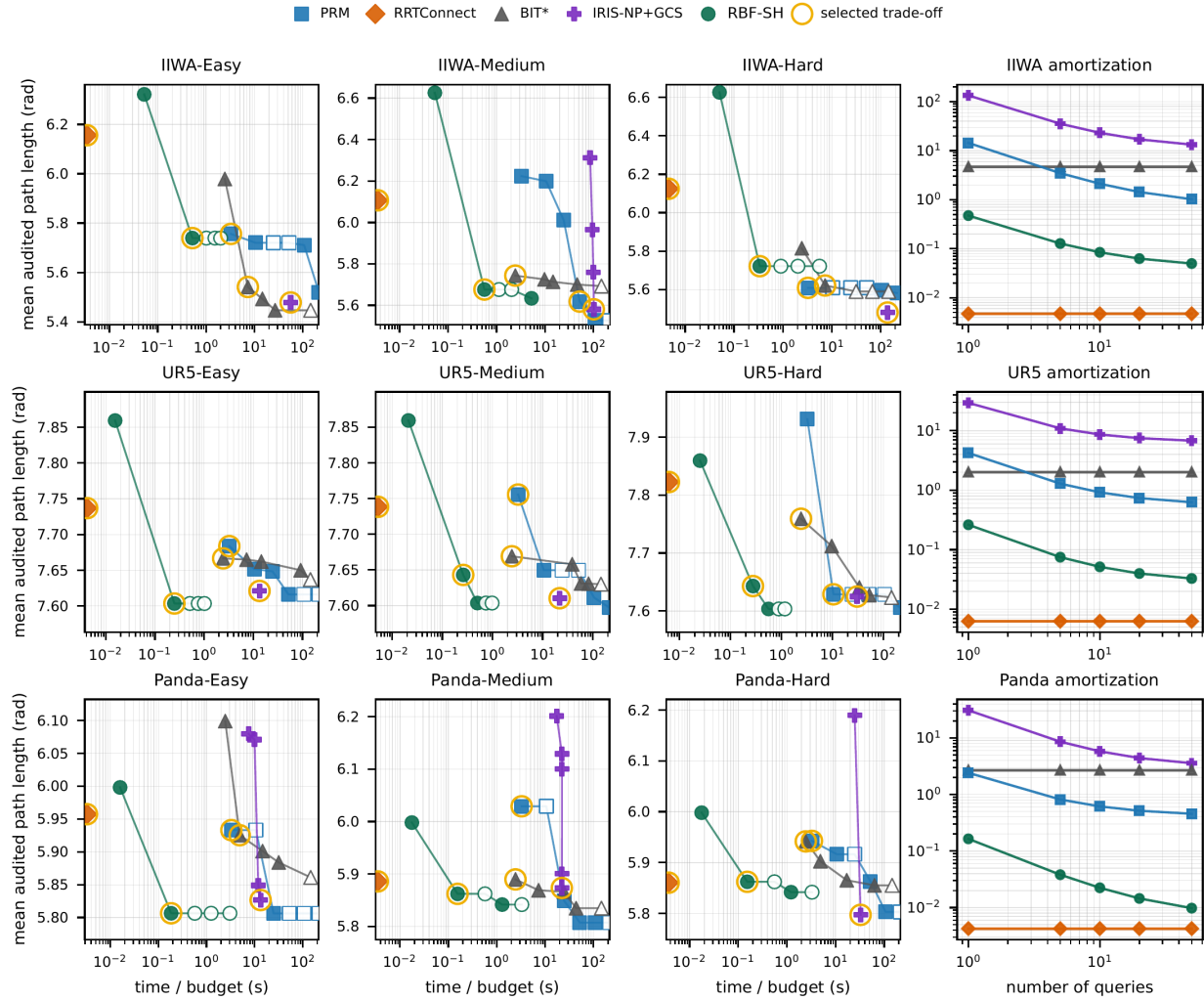


Fig. 4. Balanced random-scene final-audited design-point trade-off on IIWA, UR5, and Panda. In the trade-off panels, points nearer the lower-left corner combine lower charged time with shorter audited path length. Filled markers indicate incumbent-improving checkpoints; open markers show a sparse representative subset of nonimproving checkpoints. Gold rings mark the rows reported in table II. Small offsets separate nearly coincident IRIS-NP+GCS and PRM points.

TABLE II

Common-rule tabulated random-scene rows from Fig. 4. Gold rings identify these rows only to expose detailed numeric values; the full curves remain the primary evidence. Each method and scenario is evaluated on its full curve, and the table lists one readable slice because the complete checkpoint data are too extensive to print in full. Among points within 8% of the shortest audited path, the rule minimizes a single path/log-time utility. BIT* rows are taken from the audited monotone incumbent checkpoint trace under the same 0.01 OMPL planning/simplify and fixed-resolution final-audit segment step.

Scenario	SBF-SH			IRIS-NP+GCS			PRM			RRTConnect		BIT*	
	Build (s)	Query (s)	Path	Build (s)	Query (s)	Path	Build (s)	Query (s)	Path	Query (s)	Path	Query (s)	Path
IIWA-Easy	0.493	0.029	5.74	66.0	10.4	5.46	2.00	0.412	5.77	0.0042	6.17	6.03	5.52
IIWA-Medium	0.509	0.054	5.68	126	11.7	5.56	37.0	1.42	5.64	0.0043	6.13	2.01	5.72
IIWA-Hard	0.297	0.040	5.72	177	10.5	5.46	2.00	0.415	5.63	0.0056	6.14	6.03	5.60
UR5-Easy	0.222	0.026	7.60	12.1	6.04	7.62	2.00	0.413	7.69	0.0050	7.74	2.01	7.66
UR5-Medium	0.233	0.028	7.64	23.5	5.66	7.60	2.00	0.413	7.76	0.0063	7.74	2.01	7.66
UR5-Hard	0.246	0.031	7.64	33.4	7.27	7.62	7.00	0.832	7.64	0.0075	7.83	2.01	7.75
Panda-Easy	0.178	0.0061	5.81	15.1	3.03	5.82	2.00	0.414	5.94	0.0040	5.96	4.02	5.92
Panda-Medium	0.148	0.0074	5.86	27.1	3.02	5.87	2.00	0.416	6.04	0.0042	5.89	2.01	5.88
Panda-Hard	0.147	0.0064	5.86	41.3	3.00	5.79	2.00	0.416	5.95	0.0045	5.87	2.01	5.93

0.054 s; the corresponding PRM rows require 2.00–37.0 s charged build, and the corresponding IRIS-NP+GCS rows require 12.1–177 s. The price is a path-quality gap that varies by robot and scene: at the tabulated rows, RBF-SH remains within 0.092 rad of PRM and within 0.217 rad of BIT*, but IRIS-NP+GCS and BIT* give shorter au-

dated paths in several panels. This is the central empirical message: RBF trades very low reusable build/query cost for near-baseline, not uniformly best, audited path length. Because the IRIS seed generation differs between Shelf+IIWA and the random scenes, these timings quantify the current end-to-end pipelines rather than seed-

independent convex-region construction cost.

Against the stronger single-query reference BIT*, RBF occupies a different point in the design space: reusable build with near-baseline path quality at much smaller online planning budgets. RRTConnect serves as a no-build latency floor, either connecting within the maximum time-out or contributing a failed trial. BIT* provides the single-query quality curve through the audited monotone incumbent extracted from one long run, with the hollow markers in the figure showing a sparse representative subset of nonimproving checkpoints. This comparison shows the observed path-length/time trade-off of the implemented RBF pipeline, while leaving a broader strategy-ablation study for future work.

Accounting matches the deployment regimes: RBF contributes reusable link-envelope box-forest build, PRM reusable roadmap build, IRIS-NP+GCS convex-region build before GCS query, and RRTConnect/BIT* no reusable build phase. The remaining RBF gaps to the shortest single-query points mainly reflect the absence of a dedicated post-corridor path optimizer.

C. Mechanism Studies: Envelopes and Cache

a) *Endpoint sources.*: Table III summarizes the endpoint interval-envelope source comparison used to choose the geometric inputs for RBF-SH. The table reports endpoint-interval AABB volume, mean evaluation time, and maximum gap to the sampled reference union at each fixed width used in the standalone endpoint study. Figure 5 visualizes the same width-wise trade-off. In this updated table, the certified outer-bound baselines are IFK and two representative hierarchical variants, HIFK_3 and HIFK_5, so smaller volume is better for those rows; CritSample, Analytical, and MC estimate the sampled reference union, so larger volume is better there. All three certified AA sources remain gap-free against the sampled reference union over the reported widths. HIFK_3 already removes most of IFK’s wide-box slack at modest extra cost, and HIFK_5 pushes that trade-off further: at width 0.50 rad the mean endpoint volume drops from 5.769 m³ for IFK to 2.091 m³ for HIFK_3 and 1.401 m³ for HIFK_5, while mean endpoint cost rises from 6.1 to 25.8 to 77.1 μ s. CritSample still tracks Analytical much more closely than any of the certified AA sources on wide boxes, reaching 0.6564 versus 0.6665 m³ at 0.50 rad while remaining orders of magnitude cheaper (18.3 versus 8107.6 μ s), but it remains an advisory rather than certified bound.

b) *Link-envelope representations.*: The envelope microkernel fixes the endpoint source to CritSample and asks how the same endpoint boxes should be lifted to link-sweep envelopes. Table IV compares LinkIAABB, KDOP26, and pure SupportHull at the common short-link split $S = 4$. The table reports uninflated short-link record volume, envelope construction time, and envelope-only obstacle-check time. The representation trade-off motivates the loose-to-tight RBF-SH cascade used in the planning experiments: cheap AABB rejection handles most pairs first,

KDOP26 slabs provide a mid-phase for survivors, and SupportHull/GJK is reserved for the small set that still needs the tightest geometric witness.

c) *LECT reuse.*: Table V reports LECT as a runtime optimization rather than as the main experimental claim. Warm rows prewarm LECT on blind random-obstacle scenes and then run a fresh route-matched target build on the held-out target scene. As the random-scene prewarm budget grows, the LECT hit rate rises from 23.0% to 33.8% and the target-build median falls from 0.249 s to 0.219 s, a modest but measurable gain. Replay is the same-scene upper reference: it yields 96.9% cached-envelope hit and 2.23 $\times$ whole-build speedup. These numbers support LECT as a selective cache for revisited expensive envelope records, while the main planning conclusions come from the anytime trade-off figures above.

d) *RBF-only no-grid diagnostics.*: To isolate the lifelong-cache protocol from the broader planning pipeline, the appendix also reports a dedicated RBF-only diagnostic suite after the no-grid cleanup. Table VII shows that the d18 prewarm now charges FK/link-envelope materialization only on the leaf layer: the first pass stores 524287 evidence records while directly materializing 262144 leaf records, and the reopen pass reuses 262144 cached leaf records without fresh FK work above the leaves. On the Shelf+IIWA appendix sweep, all 6 rows pass strict audit; the best row is LinkIAABB at budget 256, with 0.0032 s build, 46.05 ms query, and 205.00 cached-evidence hits (table VIII). Under the reviewer-safe warm-cache protocol, the Shelf+IIWA warm row builds in 2.501 s with 0 external evidence hits and 2.735 ms median query time, while the balanced random-scene warm rows span 0.0016–0.5569 s build and 0.6615–3.536 ms query with at least 100.00% audit success (table IX). Rechecking the earlier UR5 audit failures showed that they came from an under-budgeted random-scene RBF runner rather than from center-to-center stitching or shortcut drift; restoring the original random-scene growth and parallelism budget removed that failure mode. The remaining IIWA warm rows are therefore best read as final-audited diagnostics rather than as reusable-corridor builds, since some scenes still succeed only through the bounded repair stage after an empty forest build.

D. Dynamic-Update Capability Study

Table VI summarizes the local update procedure on the same balanced random-scene construction used in the anytime study. This creates two matched edit paths: insertion from easy to medium to hard, and deletion from hard to medium to easy. The warm baseline is measured per transition, not as a cumulative path artifact: the source scene first populates an isolated fingerprint-cache namespace, and the reported warm time is then a fresh target-scene `build_coverage` call from that cache. This matches the random-scene warm-cache accounting and avoids carrying hidden in-memory state from earlier transitions.

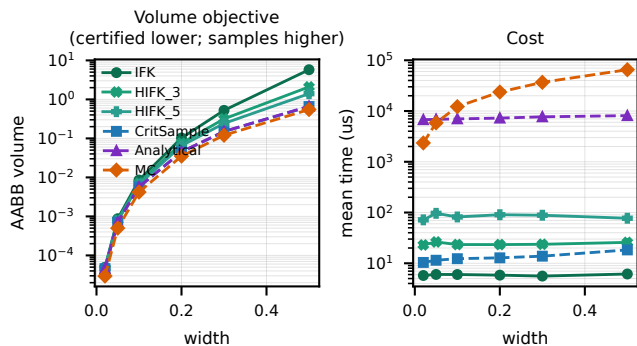


Fig. 5. Endpoint interval-envelope source study for link-envelope box validation. Left: endpoint AABB volume versus fixed width. Right: mean evaluation time versus fixed width.

TABLE III

Endpoint-interval AABB source comparison at fixed widths.

Width	Source	V (m ³)	Mean (μ s)	Max gap (m)
0.02	IFK	0.000050	5.7	0.0000
0.02	HIFK_3	0.000047	23.0	0.0000
0.02	HIFK_5	0.000047	71.7	0.0000
0.02	CritSample	0.000046	10.4	0.0000
0.02	Analytical	0.000046	6744.8	0.0000
0.02	MC	0.000029	2348.3	0.0061
0.05	IFK	0.000873	6.0	0.0000
0.05	HIFK_3	0.000801	26.3	0.0000
0.05	HIFK_5	0.000765	96.4	0.0000
0.05	CritSample	0.000713	11.5	0.0002
0.05	Analytical	0.000714	6855.4	0.0000
0.05	MC	0.000500	5850.2	0.0129
0.10	IFK	0.0085	6.0	0.0000
0.10	HIFK_3	0.0071	23.4	0.0000
0.10	HIFK_5	0.0067	81.8	0.0000
0.10	CritSample	0.0057	12.3	0.0012
0.10	Analytical	0.0057	6999.9	0.0000
0.10	MC	0.0042	12197.6	0.0277
0.20	IFK	0.1025	5.8	0.0000
0.20	HIFK_3	0.0754	23.4	0.0000
0.20	HIFK_5	0.0649	90.5	0.0000
0.20	CritSample	0.0451	12.8	0.0052
0.20	Analytical	0.0453	7249.9	0.0000
0.20	MC	0.0351	23648.7	0.0702
0.30	IFK	0.5235	5.6	0.0000
0.30	HIFK_3	0.3164	23.7	0.0000
0.30	HIFK_5	0.2419	88.5	0.0000
0.30	CritSample	0.1497	13.8	0.0126
0.30	Analytical	0.1506	7688.6	0.0000
0.30	MC	0.1204	36783.7	0.0869
0.50	IFK	5.769	6.1	0.0000
0.50	HIFK_3	2.091	25.8	0.0000
0.50	HIFK_5	1.401	77.1	0.0000
0.50	CritSample	0.6564	18.3	0.0324
0.50	Analytical	0.6665	8107.6	0.0030
0.50	MC	0.5485	65157.7	0.1122

TABLE VI

Random-scene dynamic obstacle updates on matched easy/medium/hard edit paths. Warm denotes an isolated source-scene prewarm followed by a fresh target-scene rebuild, and speedup denotes the ratio of warm median to incremental median.

Transition	Incr. (s)	Warm (s)	Speedup
easy->medium	0.0095	0.0902	9.493
medium->hard	0.0364	0.0702	1.929
hard->medium	0.0014	0.0581	41.07
medium->easy	0.000330	0.0464	140.36

The incremental update is consistently faster than this cache-warm rebuild baseline on the five-seed IIWA dynamic protocol. Insertion medians are 9.5 ms for easy→medium and 36.4 ms for medium→hard, compared with 90.2 ms and 70.2 ms for warm rebuilds. Deletion is cheaper still because previously accepted boxes remain

TABLE IV

Width-wise CritSample link-envelope comparison under the same $S = 4$ short-link split.

Width	Envelope	V (m ³)	Eval (μ s)	Collision (μ s)
0.02	LinkIAABB	0.0017	0.5	0.2
0.02	KDOP26	0.000706	1.3	0.3
0.02	SupportHull	0.000424	0.5	0.4
0.05	LinkIAABB	0.0056	0.5	0.2
0.05	KDOP26	0.0041	1.3	0.4
0.05	SupportHull	0.0035	0.5	0.5
0.10	LinkIAABB	0.0229	0.5	0.2
0.10	KDOP26	0.0205	1.3	0.4
0.10	SupportHull	0.0192	0.5	0.9
0.20	LinkIAABB	0.1255	0.5	0.2
0.20	KDOP26	0.1214	1.3	0.5
0.20	SupportHull	0.1189	0.4	0.9
0.30	LinkIAABB	0.3678	0.5	0.3
0.30	KDOP26	0.3621	1.3	0.6
0.30	SupportHull	0.3586	0.4	1.3
0.50	LinkIAABB	1.465	0.6	0.4
0.50	KDOP26	1.457	1.5	0.9
0.50	SupportHull	1.452	0.4	2.4

TABLE V

Random IIWA LECT reuse under the paper-facing CritSample endpoint source and an approximately 250-box target regime. Warm rows use blind random-obstacle prewarm scenes; Cold and Warm replay the same validation/split route. LECT hit reports external prewarm materialization hits for Warm and cached-envelope hits for Replay. Speedup charges only the target build.

Regime	Off./pop. (s)	LECT hit (%)	Build (s)	Speedup
Cold	—	0.0	0.249	1.00×
Warm	0.394	23.0	0.232	1.07×
Warm	0.758	24.7	0.235	1.06×
Warm	1.793	26.9	0.225	1.11×
Warm	3.468	28.5	0.228	1.09×
Warm	19.980	33.8	0.219	1.14×
Replay	0.322	96.9	0.111	2.23×

valid when obstacles are removed: hard→medium takes 1.4 ms and medium→easy takes 0.33 ms, versus 58.1 ms and 46.4 ms warm target rebuilds. The apparent build-time ordering is driven by transition geometry and scene topology: the target easy warm rebuild is the shortest target rebuild after cache isolation, while easy→medium is longer because it measures rebuilding the medium target from the easy source scene. Incremental audit succeeds on all insertion and medium→easy deletion cases and on 4/5 hard→medium deletions; failed audits remain failures under the same accounting used elsewhere in the paper.

VII. Conclusion

This paper presented RBF, a link-envelope-guided box-forest method for fast multi-query manipulator planning. Its geometric core maps endpoint interval envelopes to

link interval envelopes via capsule Minkowski decomposition and convex-hull generator bounding; an RRT-inspired grower builds a scene-specific box forest into a corridor graph queried by graph search and local refinement; LECT optionally caches kinematic envelope records to reduce repeated materialization cost. On the studied Shelf+IIWA and balanced IIWA/UR5/Panda workloads, the implemented RBF pipeline offers a low reusable build/query design point—below 1.1 s build and 0.11 s query—with near-baseline audited path lengths under common post-processing and fixed-resolution final audit.

a) Limitations.: The box-corridor guarantee (theorem 1) is conditional on conservative IFK validation and path containment inside the validated box union; the main reported rows do not claim that condition, and no corridor-containment fields were stored in the current artifacts. The paper-wide 663/668 audit count reflects fixed-resolution (0.01 joint-space step) empirical checks, not continuous safety guarantees. RBF is best viewed as an integration of link-envelope filtering, adaptive cell decomposition, and frontier-biased sampling; the main anytime curves are full-pipeline results that do not isolate the envelope primitive, box representation, or grower heuristics. Cross-scene LECT gains are modest at the tested prewarm budgets; the dynamic-update result is a capability study on the reported IIWA edit protocol, not a broad claim about dynamic scenes across robots or edit distributions.

b) Scope and Future Work.: The empirical scope covers one shelf workload, balanced IIWA/UR5/Panda random scenes at 50 seeds per robot/difficulty setting, and the present budget grid; distributional generalization, grower ablations, worker-count sensitivity, and narrow-clearance stress tests remain uncharacterized. Important extensions include broader scene families and larger seed sets, explicit box-containment coverage instrumentation, broader audit-step robustness checks, containment-preserving path optimization, stronger forest connectivity, and richer cache-transfer and dynamic-update benchmarks.

Appendix A

RBF-only No-grid Diagnostics

These appendix tables record the reviewer-safe RBF-only protocol used in the final reruns: d18 prewarm materializes FK/link-envelope evidence only on the leaf layer, Shelf+IIWA warm rows keep the prewarmed cache read-only, and balanced random-scene rows may backfill newly materialized online evidence into the active database.

References

- [1] R. Deits and R. Tedrake, “Computing large convex regions of obstacle-free space through semidefinite programming,” in *Algorithmic Foundations of Robotics XI*, 2015, pp. 109–124.
- [2] T. Marcucci, M. Petersen, D. von Wrangel, and R. Tedrake, “Motion planning around obstacles with convex optimization,” *Sci. Robot.*, vol. 8, no. 84, 2023.
- [3] T. Marcucci, J. Umenberger, P. Parrilo, and R. Tedrake, “Shortest paths in graphs of convex sets,” *SIAM J. Optim.*, vol. 34, no. 1, pp. 507–532, 2024.

TABLE VII

RBF-only Gate 0 and lifelong-cache mechanism status after the no-grid refactor. In E5, FK/link-envelope materialization is charged only on the target leaf layer; parent layers are reconstructed by child-hull propagation during checkpoint.

Stage	Status	Depth	Leaf FK	Nodes	Evidence	Wall s	Reuse	Cache
Gate 0	11/11	—	—	—	—	—	—	—
E5 prewarm	pass	18	262144	524287	524287	20.22	0	0
E5 reopen	pass	18	0	524287	524287	1.460	262144	0

TABLE VIII

RBF-only appendix sweep summary with all three no-grid envelope families. Rows are ranked by strict-audit success, then build plus query time, then cache footprint.

Rank	Envelope	Budget	Audit %	Build s	Query ms	Reuse	Cache	Footprint
1	LinkIAABB	256	100.00	0.0032	46.05	205	0.00004	0.00004
2	LinkIAABB	512	100.00	0.0032	46.22	205	0.00004	0.00004
3	KDOP26	512	100.00	0.0034	46.34	205	0.00004	0.00004
4	KDOP26	256	100.00	0.0034	46.68	205	0.00004	0.00004
5	SupportHull	512	100.00	0.0035	48.82	205	0.00004	0.00004
6	SupportHull	256	100.00	0.0066	46.00	205	0.00004	0.00004

TABLE IX

RBF-only warm-versus-cold baselines under the reviewer-safe lifelong-cache protocol. Shelf rows read only the prewarmed external d18 evidence and disable active-database backfill; random-scene rows may persist newly materialized online evidence to the active database.

Exp.	Scene	Mode	Backfill	Wall s	Plan s	Maint s	Query
E2	Shelf+IIWA	Cold	no	2.289	2.282	0.0075	0.0075
E2	Shelf+IIWA	Warm d18	no	2.304	2.296	0.0074	0.0074
E2	Random IIWA (easy)	Cold	yes	0.1294	0.1227	0.0067	0.0067
E2	Random IIWA (easy)	Warm d18	yes	0.1316	0.1247	0.0069	0.0069
E3	Shelf+IIWA	Cold	no	2.435	2.423	0.0123	0.0123
E3	Shelf+IIWA	Warm d18	no	2.501	2.489	0.0129	0.0129

- [4] S. M. LaValle, *Planning Algorithms*. Cambridge University Press, 2006.
- [5] —, “Rapidly-exploring random trees: A new tool for path planning,” Iowa State University, Tech. Rep. TR 98-11, 1998.
- [6] L. E. Kavraki, P. Švestka, J.-C. Latombe, and M. H. Overmars, “Probabilistic roadmaps for path planning in high-dimensional configuration spaces,” *IEEE Trans. Robot. Autom.*, vol. 12, no. 4, pp. 566–580, 1996.
- [7] S. Karaman and E. Frazzoli, “Sampling-based algorithms for optimal motion planning,” *Int. J. Robot. Res.*, vol. 30, no. 7, pp. 846–894, 2011.
- [8] S. Redon, Y. J. Kim, M. C. Lin, and D. Manocha, “Interactive and continuous collision detection for avatars in virtual environments,” in *IEEE Virtual Reality Conference*, 2004, pp. 117–124.
- [9] S. Redon, M. C. Lin, D. Manocha, and Y. J. Kim, “Fast continuous collision detection for articulated models,” *J. Comput. Inf. Sci. Eng.*, vol. 5, no. 2, pp. 126–137, 2005.
- [10] X. Zhang, S. Redon, M. Lee, and Y. J. Kim, “Continuous collision detection for articulated models using Taylor models and temporal culling,” *ACM Trans. Graph.*, vol. 26, no. 3, p. 15, 2007.
- [11] A. Amice, H. Dai, P. Werner, A. Zhang, and R. Tedrake, “Finding and optimizing certified, collision-free regions in configuration space for robot manipulators,” in *Algorithmic Foundations of Robotics XV*. Springer, 2022, pp. 328–348.
- [12] M. Petersen and R. Tedrake, “Growing convex collision-free regions in configuration space using nonlinear programming,” 2023. [Online]. Available: <https://arxiv.org/abs/2303.14737>
- [13] P. Werner, A. Amice, T. Marcucci, D. Rus, and R. Tedrake, “Approximating robot configuration spaces with few convex sets using clique covers of visibility graphs,” in *IEEE International*

- Conference on Robotics and Automation (ICRA)*, 2024, pp. 10359–10365.
- [14] D. Blackmore and M. C. Leu, “Analysis of swept volume via Lie groups and differential equations,” *Int. J. Robot. Res.*, vol. 11, no. 6, pp. 516–537, 1992.
 - [15] K. Abdel-Malek, J. Yang, D. Blackmore, and K. Joy, “Swept volumes: foundation, perspectives, and applications,” *Int. J. Shape Model.*, vol. 12, no. 1, pp. 87–127, 2006.
 - [16] S. Gottschalk, M. C. Lin, and D. Manocha, “OBBTree: A hierarchical structure for rapid interference detection,” in *ACM SIGGRAPH*, 1996.
 - [17] E. Larsen, S. Gottschalk, M. C. Lin, and D. Manocha, “Fast proximity queries with swept sphere volumes,” University of North Carolina at Chapel Hill, Tech. Rep. TR99-018, 1999.
 - [18] E. G. Gilbert, D. W. Johnson, and S. S. Keerthi, “A fast procedure for computing the distance between complex objects in three-dimensional space,” *IEEE J. Robot. Autom.*, vol. 4, no. 2, pp. 193–203, 1988.
 - [19] G. v. d. Bergen, “A fast and robust GJK implementation for collision detection of convex objects,” *J. Graph. Tools*, vol. 4, no. 2, pp. 7–25, 1999.
 - [20] J. Pan, S. Chitta, and D. Manocha, “FCL: A general purpose library for collision and proximity queries,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2012, pp. 3859–3866.
 - [21] R. E. Moore, R. B. Kearfott, and M. J. Cloud, *Introduction to Interval Analysis*. SIAM, 2009.
 - [22] L. Jaulin, M. Kieffer, O. Didrit, and E. Walter, *Applied Interval Analysis*. Springer, 2001.
 - [23] J. M. Snyder, “Interval analysis for computer graphics,” in *Proc. ACM SIGGRAPH*, 1992, pp. 121–130.
 - [24] J.-P. Merlet, “Solving the forward kinematics of a Gough-type parallel manipulator with interval analysis,” *Int. J. Robot. Res.*, vol. 23, no. 3, pp. 221–235, 2004.
 - [25] —, “Interval analysis for certified numerical solution of problems in robotics,” *Int. J. Appl. Math. Comput. Sci.*, vol. 19, no. 3, pp. 399–412, 2009.
 - [26] F. Schwarzer, M. Saha, and J.-C. Latombe, “Adaptive dynamic collision checking for single and multiple articulated robots in complex environments,” *IEEE Trans. Robot.*, vol. 21, no. 3, pp. 338–353, 2005.
 - [27] J. A. Corrales, F. A. Candelas, and F. Torres, “Safe human-robot interaction based on dynamic sphere-swept line bounding volumes,” *Robot. Comput.-Integr. Manuf.*, vol. 27, no. 1, pp. 177–185, 2011.
 - [28] R. Bohlin and L. E. Kavraki, “Path planning using lazy PRM,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 521–528.
 - [29] C. M. Dellin and S. S. Srinivasa, “A unifying formalism for shortest path problems with expensive edge evaluations via lazy best-first search over paths with edge selectors,” in *International Conference on Automated Planning and Scheduling (ICAPS)*, 2016.
 - [30] E. Schmerling, L. Janson, and M. Pavone, “Optimal sampling-based motion planning under differential constraints: The driftless case,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015.
 - [31] D. Coleman, I. Sucan, S. Chitta, and N. Correll, “Reducing the barrier to entry of complex robotic software: A MoveIt! case study,” *J. Softw. Eng. Robot.*, vol. 5, no. 1, pp. 3–16, 2014.
 - [32] T. Lozano-Pérez, “Spatial planning: A configuration space approach,” *IEEE Trans. Comput.*, vol. C-32, no. 2, pp. 108–120, 1983.
 - [33] R. A. Brooks and T. Lozano-Pérez, “A subdivision algorithm in configuration space for findpath with rotation,” *IEEE Trans. Syst. Man Cybern.*, vol. SMC-15, no. 2, pp. 224–233, 1985.
 - [34] J. J. Kuffner and S. M. LaValle, “RRT-connect: An efficient approach to single-query path planning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2000, pp. 995–1001.
 - [35] J. D. Gammell, S. S. Srinivasa, and T. D. Barfoot, “Batch informed trees (BIT*): Sampling-based optimal planning via the heuristically guided search of implicit random geometric graphs,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 3067–3074.
 - [36] J. D. Gammell, T. D. Barfoot, and S. S. Srinivasa, “Informed sampling for asymptotically optimal path planning,” *IEEE Trans. Robot.*, vol. 34, no. 4, pp. 966–984, 2018.
 - [37] L. Janson, E. Schmerling, A. Clark, and M. Pavone, “Fast marching tree: A fast marching sampling-based method for optimal motion planning in many dimensions,” *Int. J. Robot. Res.*, vol. 34, no. 7, pp. 883–921, 2015.
 - [38] I. A. Şucan, M. Moll, and L. E. Kavraki, “The open motion planning library,” *IEEE Robot. Autom. Mag.*, vol. 19, no. 4, pp. 72–82, 2012.
 - [39] M. Moll, I. A. Şucan, and L. E. Kavraki, “Benchmarking motion planning algorithms: An extensible infrastructure for analysis and visualization,” *IEEE Robot. Autom. Mag.*, vol. 22, no. 3, pp. 96–102, 2015.
 - [40] N. Ratliff, M. Zucker, J. A. Bagnell, and S. Srinivasa, “CHOMP: Gradient optimization techniques for efficient motion planning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2009, pp. 489–494.
 - [41] M. Kalakrishnan, S. Chitta, E. Theodorou, P. Pastor, and S. Schaal, “STOMP: Stochastic trajectory optimization for motion planning,” in *IEEE International Conference on Robotics and Automation (ICRA)*, 2011, pp. 4569–4574.
 - [42] J. Schulman, Y. Duan, J. Ho, A. Lee, I. Awwal, H. Bradlow, J. Pan, S. Patil, K. Goldberg, and P. Abbeel, “Motion planning with sequential convex optimization and convex collision checking,” *Int. J. Robot. Res.*, vol. 33, no. 9, pp. 1251–1270, 2014.
 - [43] M. Mukadam, J. Dong, X. Yan, F. Dellaert, and B. Boots, “Continuous-time gaussian process motion planning via probabilistic inference,” *Int. J. Robot. Res.*, vol. 37, no. 11, pp. 1319–1340, 2018.